KAVACH AI

Product Requirements Document

Full name: Karnataka AI Visualization & Analytics for Crime Hotspots
Platform title: Karnataka Crime Intelligence Command Centre
Tagline: From fragmented crime records to proactive, explainable intelligence.
Version: 1.0
Product type: Hackathon prototype
Target challenge: Datathon 2026, Challenge 02: AI-Driven Crime Analytics & Visualization Platform
Existing foundation: InsightFlow analytics platform
Repository: `vishalcoder0912/Datathon`

---

1. Executive Summary

KAVACH AI is an AI-driven crime analytics and visualization platform designed to transform fragmented crime records into actionable, explainable intelligence.

The platform will combine crime incidents, offenders, victims, locations, police stations and socioeconomic indicators to provide:

- A state-level crime command centre
- Karnataka district and police-station analysis
- Spatiotemporal crime hotspot detection
- Crime pattern and trend discovery
- Criminal network visualization
- Repeat-offender intelligence
- Explainable district risk indicators
- Anomaly and emerging-trend alerts
- Socioeconomic correlation analysis
- A grounded natural-language AI copilot

The prototype will be built by extending the existing InsightFlow repository rather than replacing its frontend, backend, analytics or AI infrastructure.

---

2. Problem Statement

Crime records are often distributed across independent spreadsheets, district-level systems and manually generated reports.

This fragmentation creates several limitations:

- State-wide patterns are difficult to identify.
- Police stations cannot easily compare trends across districts.
- Criminal relationships remain hidden in isolated records.
- Repeat offenders operating across jurisdictions are difficult to track.
- Emerging crime spikes are detected late.
- Socioeconomic relationships are not systematically analysed.
- Static reports do not support interactive investigation.
- Predictive outputs frequently lack explanations and evidence.

KAVACH AI will convert these fragmented records into a unified intelligence platform containing geographic, temporal, relational and predictive analysis.

---

3. Product Vision

Create a privacy-conscious crime-intelligence platform that helps law-enforcement analysts move from reactive reporting to proactive and evidence-based decision support.

The system must not claim to predict exact future crimes. It must provide transparent relative-risk indicators, trend projections and anomaly alerts supported by observable data.

---

4. Product Goals

4.1 Primary goals

1. Ingest crime data from CSV, Excel and JSON files.
2. Convert inconsistent source columns into a canonical crime schema.
3. Provide Karnataka district and police-station geographic analysis.
4. Identify crime hotspots using location, time, frequency and severity.
5. Detect unusual changes against historical baselines.
6. Visualize relationships among offenders, FIRs, victims, locations, vehicles and communication identifiers.
7. Identify repeat offenders and cross-district activity.
8. Produce explainable district and offender risk indicators.
9. Analyse correlations between crime and socioeconomic indicators.
10. Allow analysts to query the platform through a grounded AI copilot.
11. Export an intelligence report containing findings, evidence and limitations.
12. Demonstrate a complete end-to-end workflow using safe synthetic data.

4.2 Success goals

The prototype is successful when a user can:

1. Load the synthetic Karnataka crime dataset.
2. View state-level crime KPIs.
3. find high-risk districts.
4. open a district on the Karnataka map.
5. drill into its police stations.
6. detect an unusual crime spike.
7. inspect a criminal network.
8. identify a repeat offender.
9. view an explainable risk score.
10. ask the AI copilot to explain the finding.
11. export the result as an intelligence report.

---

5. Non-Goals

The prototype will not attempt to provide:

- Real deployment inside Karnataka State Police infrastructure
- Integration with live FIR or government databases
- Exact prediction of where or when a crime will happen
- Facial recognition
- Biometric analysis
- Automated arrest or enforcement recommendations
- A production Neo4j migration
- A production PostGIS migration
- A complete mobile application
- Full Kannada voice interaction
- Complex real-time streaming infrastructure
- Final legal or operational policing decisions
- Use of real personally identifiable crime data

All demonstration data must be fictional, synthetic or properly anonymised.

---

6. Target Users

6.1 Crime Analyst

Needs to:

- Explore crime trends
- Compare districts
- Detect anomalies
- Build intelligence reports
- Analyse socioeconomic correlations

6.2 Investigator

Needs to:

- Search FIRs and offenders
- Inspect related incidents
- Explore criminal relationships
- Review evidence references
- Identify repeat-offender patterns

6.3 Supervisor

Needs to:

- Monitor district and police-station performance
- Review active alerts
- Compare current and historical risks
- Review high-priority intelligence

6.4 Policymaker

Needs to:

- Access aggregate, anonymised trends
- Understand geographic crime distribution
- Study socioeconomic relationships
- Review strategic indicators without raw PII

6.5 Administrator

Prototype-level responsibilities:

- Load demonstration data

- Reset the demonstration environment
- Review data quality
- Access system health information

Full production role-based access control is a post-prototype enhancement.

---

7. Core Product Modules

7.1 Crime Command Centre

Purpose

Provide a state-level overview of current crime intelligence.

Required KPIs

- Total incidents
- Active investigations
- Closed investigations
- High-risk districts
- Active hotspots
- Repeat offenders
- Current anomaly alerts
- Crime change versus previous period
- Most common crime category
- Average investigation duration
- Data-quality score

Required visualizations

- Crime trend line
- Crime-category distribution
- District ranking
- Investigation-status breakdown
- Time-of-day distribution
- Alert timeline
- High-risk district list

Global filters

- Date range
- District
- Police station
- Crime category
- Crime subtype
- Investigation status

- Severity
- Time of day
- Offender status

Route

`/dashboard`

7.2 Geospatial Intelligence

Purpose

Visualize geographic crime distribution across Karnataka.

Required capabilities

- Karnataka district choropleth
- District incident counts
- District relative-risk score
- Police-station markers
- Incident markers when coordinates are available
- Hotspot visualization
- District click interaction
- District-to-police-station drilldown
- Crime category filtering
- Date filtering
- Time-of-day filtering
- Current versus previous period comparison
- District intelligence side panel
- Top police-station ranking

District panel content

- Incident count
- Crime trend
- Top categories
- Hotspot count
- Repeat-offender activity
- Active alerts
- Relative-risk score
- Supporting factors

Route

`/geo-intelligence`

7.3 Crime Trend Intelligence

Purpose

Identify temporal, seasonal and behavioural crime patterns.

Required capabilities

- Monthly trends
- Weekly trends
- Day-of-week patterns
- Hour-of-day patterns
- Morning, afternoon, evening and night comparisons
- Seasonal pattern detection
- District comparisons
- Police-station comparisons
- Crime-category growth
- Modus-operandi trends
- Current versus historical baseline
- Significant increase and decrease indicators
- Emerging crime-category alerts

Example insight

"Vehicle theft in Bengaluru Urban increased by 28% during the current 30-day period compared with the previous 30-day period."

Route

`/trend-intelligence`

7.4 Criminal Network Explorer

Purpose

Reveal hidden relationships across offenders, incidents, victims, locations and assets.

Required node types

- Offender
- Incident or FIR
- Victim
- Witness
- Location
- Police station
- Phone number

- Vehicle
- Bank account
- Digital identity
- Modus operandi

Required edge types

- `ACCUSED_IN`
- `VICTIM_IN`
- `WITNESS_IN`
- `ASSOCIATED_WITH`
- `USED_PHONE`
- `USED_VEHICLE`
- `SHARED_ADDRESS`
- `SHARED_ACCOUNT`
- `OCCURRED_AT`
- `SIMILAR_MODUS_OPERANDI`

Required capabilities

- Interactive node-and-edge graph
- Search by offender or FIR
- Filter by node type
- Filter by relationship type
- Expand connected nodes
- Highlight repeat offenders
- Display relationship strength
- Display supporting evidence
- Show node details in a side drawer
- Detect connected components or communities
- Show cross-district associations
- Limit rendering for large graphs
- Reset and fit graph controls

Recommended library

- Cytoscape.js
- `react-cytoscapejs`

Route

`/network-intelligence`

7.5 Repeat-Offender Intelligence

Purpose

Identify habitual and cross-jurisdiction offenders.

Required offender profile

- Masked offender ID
- Number of linked incidents
- Crime categories
- Active districts
- Known associates
- First incident date
- Most recent incident date
- Common modus operandi
- Associated phones
- Associated vehicles
- Associated locations
- Relative-risk score
- Risk factors
- Supporting FIR references

Candidate repeat-offender logic

An offender is marked as a repeat-offender candidate when one or more conditions are met:

- Linked incidents greater than or equal to two
- Similar modus operandi appears in two or more incidents
- Incidents occur across two or more districts
- Three or more known criminal associations
- Recent incident frequency exceeds the configured threshold

Route

`/offenders`

Detail route

`/offenders/:offenderId`

7.6 Risk and Anomaly Intelligence

Purpose

Provide transparent relative-risk indicators and identify unusual activity.

District risk formula

The prototype will use a deterministic weighted score:

- Normalized incident frequency: 25%
- Recent growth rate: 20%
- Crime severity: 15%
- Repeat-offender activity: 15%
- Hotspot density: 10%
- Criminal-network activity: 10%
- Anomaly score: 5%

Score output

Each score must include:

- Score from 0 to 100
- Risk band
- Calculation timestamp
- Contributing factors
- Data period
- Confidence level
- Model or formula version
- Known limitations
- Decision-support disclaimer

Risk bands

- 0–29: Low
- 30–49: Moderate
- 50–69: Elevated
- 70–84: High
- 85–100: Critical

Anomaly categories

- District incident spike
- Police-station incident spike
- Crime-category spike
- Time-of-day anomaly
- Modus-operandi anomaly
- Repeat-offender activity increase
- New criminal association
- Investigation-delay anomaly

Required methods

Prototype analytics may use:

- Z-score
- Interquartile range
- Rolling averages
- Period-over-period percentage change
- Deterministic weighted scoring

Route

`/risk-intelligence`

7.7 Socioeconomic Crime Intelligence

Purpose

Study statistical relationships between district-level social indicators and crime.

Required indicators

- Population density
- Urbanisation rate
- Literacy rate
- Unemployment rate
- Migration rate
- Average income
- Education index
- Housing density
- Economic stress indicator

Required visualizations

- Correlation matrix
- Scatter plot
- Bubble chart
- District comparison
- Ranked correlations
- Crime-rate overlay

Required output

Every result must display:

- Correlation coefficient

- Direction
- Strength
- Sample size
- Data period
- Limitations
- Correlation-versus-causation notice

Mandatory notice

“Statistical correlation indicates association and does not prove causation.”

Route

`/social-intelligence`

7.8 AI Crime Intelligence Copilot

Purpose

Enable natural-language exploration of the analytics platform.

Example questions

- Show vehicle-theft hotspots during the last six months.
- Which districts recorded unusual cybercrime growth?
- List repeat offenders active across multiple districts.
- Find relationships connected to FIR-2026-0041.
- Compare robbery incidents during daytime and night-time.
- Explain why Mysuru received an elevated risk score.
- Create a chart comparing cybercrime across districts.
- Show the strongest socioeconomic correlations.

Supported responses

- Text explanation
- KPI
- Chart
- Table
- Map filter
- District selection
- Network selection
- Offender profile
- Applied filters
- Query plan
- Supporting record references
- Confidence

- Limitations

AI safety requirements

- External LLM providers must not receive raw unrestricted rows.
- PII-like fields must be masked or excluded.
- Statistical calculations must be performed locally.
- Risk scores must be calculated deterministically.
- The LLM may explain results but must not invent records.
- Every result must reference real records or aggregates.
- Unsupported questions must produce an explicit limitation.
- The assistant must not recommend arrest, detention or enforcement actions.

Provider order

1. Ollama local model
2. Gemini when configured
3. OpenAI or Anthropic when configured
4. Deterministic local fallback

Route

`/ai-copilot`

7.9 Data Management

Purpose

Load, validate and inspect crime datasets.

Required capabilities

- CSV upload
- Excel upload
- JSON upload
- Synthetic demo dataset
- Column mapping preview
- Schema validation
- Data-quality warnings
- Invalid-row reporting
- Dataset reset
- Dataset export
- Sample row preview
- Canonical schema conversion

Route

`/data-management`

7.10 Alerts and Reports

Alert capabilities

- List active alerts
- Filter alerts
- Mark alert as reviewed
- View supporting evidence
- Show alert severity
- Show creation timestamp
- Show affected district or station

Report capabilities

Generate a downloadable intelligence report containing:

- Executive summary
- Applied filters
- KPIs
- Key trends
- Hotspots
- Active alerts
- Repeat offenders
- Risk scores
- Network findings
- Methodology
- Limitations
- Generation timestamp

Preferred export

- PDF when reliable
- HTML fallback
- CSV for raw aggregate tables

Routes

- `/alerts`
- `/reports`

---

8. Canonical Data Model

8.1 Incidents

```
incidents
- id
- fir_number
- crime_type
- crime_subtype
- description
- incident_date
- incident_time
- district_id
- police_station_id
- latitude
- longitude
- severity
- investigation_status
- modus_operandi
- created_at
- updated_at
```

8.2 Districts

```
districts
- id
- name
- code
- geojson_name
- population
- area_sq_km
```

8.3 Police stations

```
police_stations
- id
- name
- district_id
- latitude
- longitude
```

8.4 Persons

```
persons
- id
- masked_identifier
- age_group
- gender
- occupation
- socioeconomic_group
- home_district_id
```

8.5 Incident-person relationships

```
incident_persons
- incident_id
- person_id
- role
```

Supported roles:

- `ACCUSED`
- `VICTIM`
- `WITNESS`
- `COMPLAINANT`

8.6 Offender profiles

```
offender_profiles
- person_id
- linked_incident_count
- repeat_offender_status
- active_district_count
- known_associate_count
- first_incident_date
- latest_incident_date
- calculated_risk_score
```

8.7 Relationships

```
relationships
- id
- source_entity_id
```

```
- source_entity_type
- target_entity_id
- target_entity_type
- relationship_type
- strength
- evidence_reference
- created_at
```

8.8 District indicators

```
district_indicators
- district_id
- population_density
- urbanization_rate
- literacy_rate
- unemployment_rate
- migration_rate
- average_income
- education_index
- housing_density
- economic_stress_index
- reference_year
```

8.9 Alerts

```
alerts
- id
- alert_type
- district_id
- police_station_id
- crime_type
- severity
- title
- message
- evidence_json
- detected_at
- status
```

8.10 Audit events

```
audit_events
- id
- actor
```

```
- action
- entity_type
- entity_id
- metadata_json
- created_at
```

---

9. Crime Schema Adapter

The ingestion layer must map alternative source names to canonical fields.

Examples:

```
FIR_No, firNumber, FIRNumber, case_id
→ fir_number

Crime_Category, offence_type, crimeType
→ crime_type

Police_Station, PS_Name, station
→ police_station

District_Name, districtName
→ district

Accused_ID, offender_id, criminal_id
→ offender_id

Incident_Date, occurrence_date
→ incident_date

Latitude, lat
→ latitude

Longitude, lng, lon
→ longitude
```

The interface must display mapping confidence and allow corrections before import.

---

10. API Requirements

Use the `/api/kavach` namespace.

System

```
GET /api/health
GET /api/kavach/status
```

Data

```
POST /api/kavach/datasets/import
POST /api/kavach/datasets/demo
GET /api/kavach/datasets/current
DELETE /api/kavach/datasets/current
GET /api/kavach/datasets/quality
```

Dashboard

```
GET /api/kavach/overview
GET /api/kavach/kpis
```

Incidents

```
GET /api/kavach/incidents
GET /api/kavach/incidents/:incidentId
```

Geospatial analysis

```
GET /api/kavach/geo/districts
GET /api/kavach/geo/districts/:districtId
GET /api/kavach/geo/hotspots
GET /api/kavach/geo/police-stations
```

Trends

```
GET /api/kavach/trends
GET /api/kavach/trends/categories
GET /api/kavach/trends/time-of-day
GET /api/kavach/trends/districts
```

Networks

```
GET /api/kavach/network
GET /api/kavach/network/entity/:entityId
GET /api/kavach/network/offender/:offenderId
```

Offenders

```
GET /api/kavach/offenders
GET /api/kavach/offenders/:offenderId
```

Risk and anomalies

```
GET /api/kavach/risk/districts
GET /api/kavach/risk/districts/:districtId
GET /api/kavach/anomalies
```

Socioeconomic analysis

```
GET /api/kavach/social/correlations
GET /api/kavach/social/districts/:districtId
```

Alerts

```
GET /api/kavach/alerts
PATCH /api/kavach/alerts/:alertId
```

Copilot

```
POST /api/kavach/copilot/query
GET /api/kavach/copilot/history
DELETE /api/kavach/copilot/history
```

Reports

```
POST /api/kavach/reports/intelligence
GET /api/kavach/reports/:reportId
```

All list endpoints must support applicable query parameters for:

- Date range
- District
- Police station
- Crime category
- Severity
- Investigation status
- Pagination
- Sorting

---

11. Technology Stack

11.1 Prototype frontend

- React 18
- TypeScript
- Vite
- Tailwind CSS
- Radix UI
- TanStack Query
- Recharts
- React Simple Maps
- D3 Scale
- Cytoscape.js
- Framer Motion
- React Hook Form
- Zod

11.2 Prototype backend

- Existing Node.js HTTP backend
- Existing JavaScript modules
- TypeScript only for new modules when it does not disrupt the build
- Zod request validation
- Existing SQLite persistence
- Existing shared analytics package
- Existing AI provider abstraction

- PDFKit or existing export utilities
- Structured application logging

Do not rewrite the backend in NestJS or Fastify for the prototype.

11.3 Optional ML service

Use the existing Python ML service only when it already runs reliably.

Potential libraries:

- FastAPI
- pandas
- NumPy
- scikit-learn
- NetworkX
- GeoPandas
- Shapely

The first working prototype may implement hotspot, risk and anomaly calculations in Node.js to avoid creating unnecessary deployment dependencies.

11.4 Production target architecture

- PostgreSQL
- PostGIS
- Neo4j
- Redis
- Background jobs
- Strong role-based access
- Managed secrets
- Encrypted storage
- Central audit service

The production architecture must be documented but does not need to be implemented in the hackathon prototype.

---

12. UI and Design Requirements

12.1 Design direction

The interface should look like a credible government intelligence platform.

Avoid:

- Cyberpunk aesthetics
- Excessive gradients
- Neon backgrounds
- Unnecessary animation
- Decorative 3D elements
- Tiny unreadable charts

12.2 Colour system

```
Primary navy:       #0F172A
Secondary blue:     #1D4ED8
Intelligence cyan: #0891B2
Alert amber:        #D97706
Critical red:       #DC2626
Success green:      #15803D
Surface:            #F8FAFC
Card:               #FFFFFF
Primary text:       #0F172A
Muted text:         #64748B
Border:             #E2E8F0
```

12.3 Required layout

Desktop:

- Left navigation
- Top header
- Global filters
- Main analysis workspace
- Optional right-side intelligence panel

Mobile:

- Collapsible navigation
- Stacked cards
- Horizontally scrollable tables
- Simplified map and graph controls

12.4 Accessibility

- Keyboard navigation
- Visible focus states
- Semantic headings
- ARIA labels

- Sufficient colour contrast
- Text alternatives
- No information communicated by colour alone

---

13. Synthetic Demonstration Dataset

The repository must contain a safe synthetic dataset with enough complexity to demonstrate all modules.

Minimum dataset characteristics

- Multiple Karnataka districts
- Multiple police stations per selected district
- At least 1,000 synthetic incidents
- Multiple crime categories
- At least twelve months of dates
- Incident times covering all dayparts
- Severity levels
- Investigation statuses
- Latitude and longitude
- Multiple offenders linked to incidents
- Repeat offenders
- Shared phone numbers
- Shared vehicles
- Shared locations
- Similar modus operandi
- Victim records
- District socioeconomic indicators
- At least three intentional anomalies
- At least three geographic hotspots
- At least two cross-district offender networks

Required intentional demo patterns

1. Cybercrime spike in one district.
2. Vehicle-theft hotspot near selected transit areas.
3. Repeat offender active across districts.
4. Shared vehicle and phone connection among several offenders.
5. Night-time burglary pattern.
6. Socioeconomic correlation suitable for demonstration.
7. Investigation-delay anomaly.
8. One low-data district showing limited confidence.

All person identifiers must be fictional or masked.

---

14. Functional Acceptance Criteria

Dashboard

- Loads without console errors.
- KPIs reflect active filters.
- Charts use real aggregates from the loaded dataset.
- Empty states are clear.
- Loading and failure states are visible.

Map

- Karnataka district boundaries render correctly.
- Clicking a district updates its details.
- Filters update the map.
- Hotspots use calculated values.
- Police-station drilldown works.

Trends

- Period comparisons are correct.
- Day-of-week and time-of-day analysis work.
- Anomalies reference actual aggregates.
- Historical baseline logic is documented.

Network

- Nodes and edges come from loaded data.
- Clicking a node displays details.
- Search and filtering work.
- Repeat offenders are visually distinct.
- Supporting evidence is shown.

Offenders

- Repeat-offender classification is deterministic.
- Profiles show linked incidents.
- Risk factors are visible.
- Cross-district activity is displayed.

Risk

- Scores range from 0 to 100.
- Each score shows contributing factors.
- Scores update when filters change.

- A decision-support disclaimer is visible.
- No score is generated without adequate data.

Socioeconomic analysis

- Correlations use district-level aggregates.
- Sample size is displayed.
- Correlation-versus-causation warning is visible.

Copilot

- Answers are grounded in available data.
- Unsupported requests fail safely.
- The assistant can return charts or tables.
- The assistant can explain risk factors.
- The assistant does not invent FIRs or offenders.
- Raw sensitive records are not sent to external providers.

Reports

- Report generation succeeds.
- Report contains applied filters and timestamp.
- Report includes methodology and limitations.
- Exported report opens correctly.

---

15. Non-Functional Requirements

Performance

- Initial frontend load should remain usable on a normal laptop.
- Dashboard interactions should normally respond within one second after data has loaded.
- Typical API analytics requests should complete within three seconds.
- Graph rendering must be limited or paginated for large networks.
- Large chart modules must be lazy-loaded.
- The map must not rerender unnecessarily.

Reliability

- No unhandled promise rejections.
- No fatal console errors.
- Backend must return consistent error envelopes.
- The demo dataset must load reliably.
- The application must survive AI-provider failure through deterministic fallbacks.

Security

- No secrets committed to Git.
- Environment variables documented in `.env.example`.
- PII must be masked.
- Inputs must be validated.
- SQL injection must be prevented.
- External AI requests must use safe schema or aggregate context.
- Reports must not expose hidden raw PII.

Maintainability

- New KAVACH modules must be separated from unrelated legacy code.
- Shared crime types must be centralized.
- Analytics formulas must include tests.
- Major algorithms must include concise documentation.
- No duplicate source-of-truth calculations across frontend and backend.

---

16. Testing Requirements

Unit tests

Required for:

- Crime schema mapping
- District aggregation
- Time-of-day classification
- Trend calculations
- Period comparison
- Hotspot score
- District risk score
- Repeat-offender classification
- Relationship generation
- Anomaly detection
- PII masking

API tests

Required for:

- Demo dataset loading
- Overview endpoint
- District endpoint
- Hotspot endpoint

- Network endpoint
- Offender endpoint
- Risk endpoint
- Copilot endpoint
- Report generation

Frontend tests

Required for:

- Dashboard loading
- Filter updates
- District selection
- Network node selection
- Offender profile rendering
- Risk explanation rendering
- Copilot response rendering
- Empty states
- API failure states

End-to-end test

One Playwright test must perform:

1. Load demo dataset.
2. Open dashboard.
3. Filter a crime category.
4. Open geographic analysis.
5. Select a district.
6. Open network intelligence.
7. Select a repeat offender.
8. Open AI copilot.
9. ask for an explanation.
10. Generate a report.

---

17. Deployment Requirements

Prototype deployment

Preferred:

- Frontend: Vercel
- Node backend: GCP Cloud Run, Compute Engine or another stable Node host
- SQLite: persistent backend volume where supported

- AI: deterministic fallback by default
- Optional Ollama: local demonstration environment

If persistent SQLite storage is not supported in the selected cloud environment, bundle the synthetic dataset and rebuild the demonstration database at startup.

Environment variables

```
NODE_ENV
PORT
FRONTEND_ORIGIN
DATABASE_PATH
GOOGLE_API_KEY
OPENAI_API_KEY
ANTHROPIC_API_KEY
OLLAMA_BASE_URL
AI_PROVIDER
REPORT_OUTPUT_DIR
```

No AI key should be required to run the core prototype.

---

18. Suggested Repository Structure

```
apps/
├── frontend/
│   └── src/
│       ├── app/
│       ├── features/
│       │   ├── command-center/
│       │   ├── geo-intelligence/
│       │   ├── crime-trends/
│       │   ├── criminal-network/
│       │   ├── offenders/
│       │   ├── risk-intelligence/
│       │   ├── socioeconomic/
│       │   ├── ai-copilot/
│       │   ├── alerts/
│       │   └── reports/
│       └── shared/
│
├── backend/
│   └── src/
│       ├── modules/
```

```
│       │    ├── crime/
│       │    ├── geo/
│       │    ├── trends/
│       │    ├── network/
│       │    ├── offenders/
│       │    ├── risk/
│       │    ├── socioeconomic/
│       │    ├── alerts/
│       │    ├── reports/
│       │    └── copilot/
│       ├── database/
│       ├── middleware/
│       └── services/
│
└── ml-service/

packages/
├── shared-types/
├── shared-analytics/
└── crime-schema/

data/
├── demo/
├── geojson/
└── socioeconomic/

docs/
├── KAVACH_AI_PRD.md
├── KAVACH_AI_ARCHITECTURE.md
├── KAVACH_AI_DEMO.md
└── KAVACH_AI_API.md
```

---

19. Implementation Priority

Phase 1: Foundation

- Create KAVACH branch
- Preserve existing working functionality
- Add KAVACH branding
- Add canonical crime types
- Add synthetic demonstration dataset
- Add schema adapter
- Add KAVACH API namespace

Phase 2: Core intelligence

- Command Centre
- Karnataka district map
- Police-station drilldown
- Trend analysis
- Hotspot calculation
- Anomaly alerts

Phase 3: Relational intelligence

- Criminal network graph
- Relationship generation
- Repeat-offender classification
- Offender detail profiles

Phase 4: Explainable intelligence

- District risk scores
- Risk-factor explanations
- Socioeconomic correlations
- AI Crime Intelligence Copilot

Phase 5: Submission hardening

- Report generation
- Unit and API tests
- Playwright flow
- Accessibility fixes
- Performance fixes
- Deployment configuration
- Demo documentation

---

20. Demonstration Script

1. Open KAVACH AI.
2. Load the synthetic Karnataka crime dataset.
3. Show state-level KPIs.
4. Highlight a cybercrime increase.
5. Open Karnataka geospatial intelligence.
6. Select a high-risk district.
7. Drill down to a police station.
8. Show a calculated hotspot.
9. Open Criminal Network Explorer.
10. Select a repeat offender.

11. Show linked incidents, phone and vehicle.
12. Open Risk Intelligence.
13. Explain the offender or district score.
14. Ask the AI copilot why the risk increased.
15. Show evidence and methodology.
16. Export the intelligence report.

---

21. Definition of Done

The prototype is complete when:

- All required routes are accessible.
- The demonstration dataset loads successfully.
- The Command Centre shows real calculated results.
- Karnataka geographic analysis works.
- Hotspots and anomalies are calculated.
- The criminal network is interactive.
- Repeat offenders are identified.
- Risk scores are explainable.
- Socioeconomic correlations are displayed responsibly.
- The AI copilot is grounded and has a local fallback.
- Intelligence report export works.
- Core tests pass.
- The build succeeds.
- The primary Playwright demonstration flow passes.
- No fatal browser-console errors remain.
- Setup and deployment instructions are documented.
- The application can be demonstrated without external AI credentials.

---

22. Product Disclaimer

KAVACH AI is a decision-support prototype. It does not determine guilt, predict exact criminal actions or replace investigative judgment. Its outputs must be interpreted with human oversight, supporting evidence, legal safeguards and awareness of data limitations.